Payton Pei

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Summary

Systems Science and Engineering master's student at the University of Ottawa, graduating December 2026. Software engineer building and fine-tuning AI systems with Python, PyTorch, Rust/Tauri and TypeScript/React; published LifeMint on the App Store. Former Nokia and Ford intern seeking new-grad AI and developer-tools roles in Ottawa.

Education

University of Ottawa
Master of Systems Science and Engineering — *2024 — Dec 2026 (expected)*

Carleton University
Bachelor of Computer Science — *2017 — 2023*

Projects

GaokaoTutor — LLM Fine-Tuning & Evaluation
- Fine-tuned Spark X2.5-4B with 4-bit QLoRA on 161 tutoring traces with data provenance and contamination checks, cutting unsolicited solution dumps from 100% to 13.5% on a frozen 80-case holdout.
- Moved to Qwen2.5-Math-1.5B with an MLX LoRA: three-vote solutions score 454/497 on held-out GAOKAO-Bench math, beating Qwen3.5-4B (449) at 58% of its compute, in a Python/Expo app with deterministic step checks.

pipi-shrimp-agent github.com/mammut001/pipi-shrimp-agent
- Built a local-first desktop AI agent with Rust/Tauri, React/TypeScript and SQLite, integrating multiple LLM providers, structured tool calls and project memory.
- Implemented provider adapters, buffered SSE parsing and cancellation, plus CDP browser automation with failure recovery and context compression for long-running sessions.

Conveyor github.com/mammut001/Conveyor
- Built a Python service linking Telegram and Feishu to Codex CLI, isolating each task in its own Git worktree.
- Added a SQLite task queue with restart recovery, operator allowlists, approval gates and audit logs.

Réussir en français — TCF Canada Prep Platform french.paytonpei.top
- Grew an open-source typing-trainer fork into a full-stack TCF Canada prep platform with A1–B2 vocabulary drills, a 26-week adaptive study plan with spaced review and timed mock exams in all four skills.
- Added calibrated LLM writing and speaking feedback with swappable providers, OAuth and passkey sign-in, Stripe and Alipay billing, and a self-hosted Next.js and Docker deployment.

LifeMint (formerly Focus Mint) apps.apple.com/us/app/lifemint/id6759029810
- Published a SwiftUI app for iPhone, iPad and Apple Watch with focus sessions, earnings tracking and 3 languages.
- Built StoreKit purchases, widgets, Live Activities and two-way iPhone–Watch sync with WatchConnectivity.

Work Experience

Teaching Assistant 2021
Carleton University — *Ottawa, Canada*
- Teaching assistant for Python 3 (2021) and JOUR 4401 (2021): relational databases, web crawling and data analysis.

Software Tester Intern Sep 2020 — Aug 2021
Nokia Canada — *Ottawa, Canada*
- Tested the legacy NFM-P client for the 7750 Service Router with JavaScript, XML and Mocha.
- Documented system architecture and data flows in Confluence.

Software Engineer Intern May 2020 — Aug 2020
Ford Motor Canada — *Ottawa, Canada*
- Wrote C++11 vehicle-cloud connectivity code on QNX/Linux with GoogleTest unit tests, bug fixes and Jenkins CI/CD.

Research

Low-Light RAW Image Restoration with Bridge U-Net and DDBM 2026
- Built the initial DDBM implementation in a collaborative low-light RAW restoration study on SID and ELD data.
- The project evaluated a DDBM-Feature variant at 26.12 dB PSNR and 0.807 MS-SSIM on 598 SID Sony test images.

Skills

- **Languages**: Python, TypeScript, JavaScript, Rust, Swift, C++
- **Frameworks**: React, Node.js, Tauri 2, SwiftUI
- **AI & Systems**: PyTorch, Transformers, PEFT/TRL, QLoRA, MLX, LLM APIs, Tool Calling, MCP, SQLite
- **Testing & Delivery**: PyTest, Vitest, Playwright, GoogleTest, Jenkins, Docker
- **Additional**: CDP Browser Automation, WatchConnectivity, StoreKit 2